

Data Manipulation and Visualisation 600



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Qualification: BSc IT Year of Study: 2 Semester: 1

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document.

Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents

Introduction

Main Body (with relevant subheadings)

Conclusion

References

Submit the assignment in PDF format on Moodle.

Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: 

Date: 15/03/26

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Requirements

Pycharm 2026.1

Matplotlib

Numpy

Seaborn

Pandas

Dash (Dash, dcc, html, Input, Output)

dash_bootstrap_components

plotly. Express

plotly.graph_objects

base64

BytesIO

NOTE !!!! PLSE MAKE SURE THE LATEST ARE DOWNLOADED TO RUN MY PROJECT

TASK ONE :

1.1 data loading

i was able to load the superstore dataset into a pandas dataframe.the dataset has information about sales transactions such as product details,customer segments,regional information and financial metrics like sales,discounts and profits.To make sure the dataset was loaded correctly and to get a sense on how the data was structured,the first 10 and last five rows were shown.i also looked at the shape of the dataset to see how many rows and columns it had.

1.2 intial exploration

to comprehend the dataset structure and data quality,i carried out a preliminary investigation.there are both numerical and category variables in the dataset.Sales,quantity,discount,and profit are important number columns.Category,sub-category,region,segment and ship mode are important category columns.While the categorical columns display several distinct values that represent several business factors ,summary stats indicates the distribution of numerical values.

the distinct values that were found are ship mode,consumer/corporate/home office segments,regions and categories.

1.3 business questions formulations

1. which product subcategories routinely perform poorly in terms of profitability and what elements like discounts or geographical locations contribute to this?

finidng product subcategories that have a detremental effect on overall profitablity is the goal of this enquiry.it is feasible to ascertain if bad performance is caused by pricing tactics,geographic considerations or product related concerns by analysing earnings in conjunction with variables like discount levels and regional performance

2.which client segments and geographical areas have the lowest profitability,and how does profitability change amongst them?

finding these variations in performance between geographical areas and client catigoires is

the main goal of this query. when sales and profits are analysed together, trends where some locations or segments produce high sales but poor profit can be found, pointing to inefficiencies or areas in need of strategic improvement.

3. how does discount levels and profitability relate to each other and when do discounts start to have a negative impact on profit?

this enquiry looks into how discounting affects profitability. it is feasible to ascertain whether larger discounts result in lower profit margins and to pinpoint any threshold at which discounts start to seriously impair profitability by examining the relationship between discount and profit.

TASK 2 :

2.1 analysis of missing values

to evaluate the completeness of the data,i performed a missing value analysis.my findings revealed that 2510 missing values or roughly 25.12% of the dataset were present in a number of columns.this suggests that a sizeable percentage of the dataset has incomplete records,most likely as a result of problems with data entry or import.i used a heat map for visualising the pattern of missing values in the dataset.in order to maintain the dataset for additional analysis, i made sure no rows were eliminated or imputed at this point despite the existence of missing values.

2.2 identifying duplicates

to guarantee data integrity,i examined duplicate rows in every column.there were no exact duplicate rows in the dataset.after i eliminated the duplicates,the dataset had 9994 rows.since several items can belong to the same order in a transactional dataset,it is not surprising that i found duplicate order id's (5739).duplicate order id's were kept as legitimate data.

2.3 conversion of data types

to facilitate precise analysis,i standardised the following data categories:

- i converted the order date and ship date to date time format which was successful.
- i transformed sales,quantity,discount and profit to numerical format which were among the numerical columns.
- i created a new column called shipping_dates to gauge delivery time.
- the formula $(\text{profit}/\text{sales}) \times 100$ was used to create a new column called profit_margin.
- from the order date,i extracted 2 further columns : order year and order month.

these changes i made enhanced the consistency of the data and made additional analysis possible.

2.4 identifying and handling outliers

the IQR technique was used to identify outliers

- sales

Q1 is 16.27

Q3 is 223.92

IQR is 207.65

therefore, 903 outliers were found.

-earnings

Q1 is 60.22

Q3 is 76.64

IQR is 16.42

therefore, 2 outliers were found

-discount

Q1 is 0.00

Q3 is 0.20

IQR is 0.20

therefore, 618 outliers were found.

i used boxplots to visualise the distribution of these variables. since the outliers most likely reflect actual business situations like this, there were left in place. high-value exchanges, purchases in bulk and strategies for high discounts, these values were kept since eliminating them would distort the analysis.

TASK 3 :

3.1 numpy's descriptive stats

i used numpy to calculate sales and profit columns (statistical measurements).

- sales

average : 246.99

median : 54.58

standard deviation : 674.40

the sales distribution is right skewed since the mean is much greater than the median.this implies that a few high-value transactions raise the average whereas the majority of transactions are rather little.

-Earnings

average: 68.12

median: 76.64

standard deviation: 24.81

the profit distribution is left skewed since the mean is less than the median which implies that lower profit values cause the average to decline.

3.2 analysis in groups

-analysis of categories

the largest profit was made by technology(408.74).office supplies and furnintures showed little profit,suggesting possible problems in that area.this implies that the distribtuion of profitability among product cantegories is not uniformed.

-analysis of regions and segments

the majority of region-segment combinations showed little profit.a few profitable ventures were consumer central (102.19) and home office in the south (131.38).this suggest uneven profitablity across the geographical areas and clientele groups.

-profit margin by subcategory

the only category with a legitimate profit margin of 21.46% was accessories. NaN values, which indicate incomplete or missing profit data were returned by the majority of other subcategories.

-trends in monthly sales

over time, sales varied with prominent peaks like november 2014(67195.30) and september 2014 (64930.40). this implies that sales performance varies seasonally.

3.3 analysis of correlation

i examined the associations between important variables using correlation matrix. sales and profit have the strongest positive correlation (0.71). better sales are typically linked to better profit. profit and discount have the strongest negative association(-0.61). higher discounts considerably lower profitability. this suggests that profit margins are significantly impacted negatively by bargain tactics.

3.4 analysis of profitability

to find transactions that were losing money, i filtered the dataset.

total amount lost: 0.00

there were zero transactions that result in a loss. the dataset contained no trades that resulted in losses. given the nature of retail data, this is unusual and could be the result of missing profit figures found during the cleaning process or problems with data quality.

TASK 4

PICTURES OF THE DASHBBOARD

SuperStore Performance & Profitability Dashboard

Student Number: 402309260

Interactive dashboard analysing sales, profit, discount impact, regional performance, shipping efficiency, and customer segments.

Filters

Region

Select region(s)

Category

Select category(s)

Segment

Select segment(s)

Date Range

2014-01-04 → 2017-12-30

Regional Chart Metric

☒ Sales ☐ Profit

Total Sales

\$1,848,453.8
5

Total Profit

\$408.74

Average Profit Margin

21.46%

Number of Orders

4,254

Overview

Regional Analysis

Product & Segment Analysis

Static Visualisations

Sales vs Profit Scatter Plot

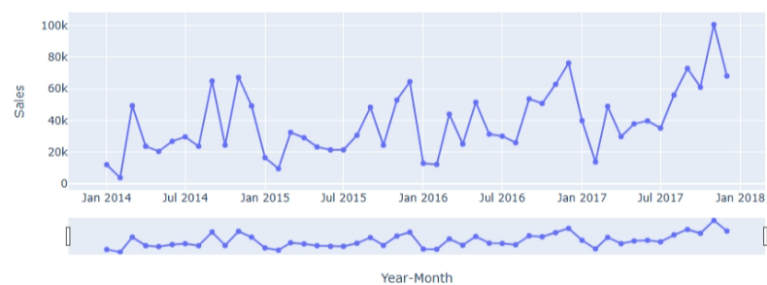
Sales vs Profit



This chart shows the relationship between sales and profit. Marker size reflects quantity and colour reflects customer segment.

Monthly Trend

Monthly Sales Trend



The time series shows sales over time with an interactive range slider and preset buttons.

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Segment

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Date Range

2014-01-04 → 2017-12-30

Regional Chart Metric

☒ Sales ☐ Profit

Total Sales

\$1,848,453.85

Total Profit

\$408.74

Average Profit Margin

21.46%

Number of Orders

4,254

Overview

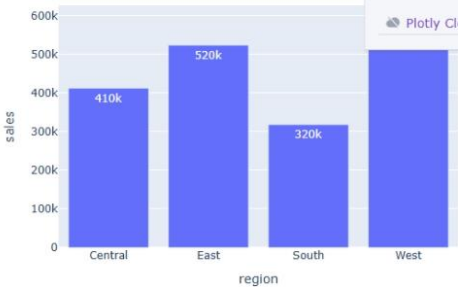
Regional Analysis

Product & Segment Analysis

Static Visualisations

Regional Performance

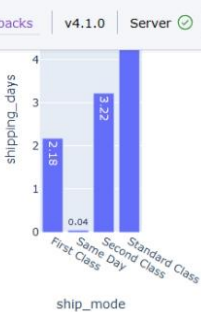
Regional Sales Comparison



This chart compares regional performance using the selected metric from the radio buttons.

Shipping Efficiency

Average Shipping Days by Ship



Average shipping days by ship mode help identify delivery efficiency differences.

SuperStore Performance & Profitability Dashboard

Student Number: 402309260

Interactive dashboard analysing sales, profit, discount impact, regional performance, shipping efficiency, and customer segments.

Filters

Region

Select region(s)

Category

Select category(s)

Segment

Select segment(s)

Date Range

2014-01-04 → 2017-12-30

Regional Chart Metric

☒ Sales ☐ Profit

Total Sales

\$1,848,453.85

Total Profit

\$408.74

Average Profit Margin

21.46%

Number of Orders

4,254

Overview

Regional Analysis

Product & Segment Analysis

Static Visualisations

Segment Performance

Sales and Profit by Segment



This chart compares total sales and total profit across customer segments.

SuperStore Performance & Profitability Dashboard

Student Number: 402309260

Interactive dashboard analysing sales, profit, discount impact, regional performance, shipping efficiency, and customer segments.

Filters

Region

Select region(s)

Category

Select category(s)

Segment

Select segment(s)

Date Range

2014-01-04 → 2017-12-30

Regional Chart Metric

☒ Sales ☐ Profit

Total Sales

\$1,848,453.85

Total Profit

\$408.74

Average Profit Margin

21.46%

Number of Orders

4,254

Overview

Regional Analysis

Product & Segment Analysis

Static Visualisations

Profit Distribution



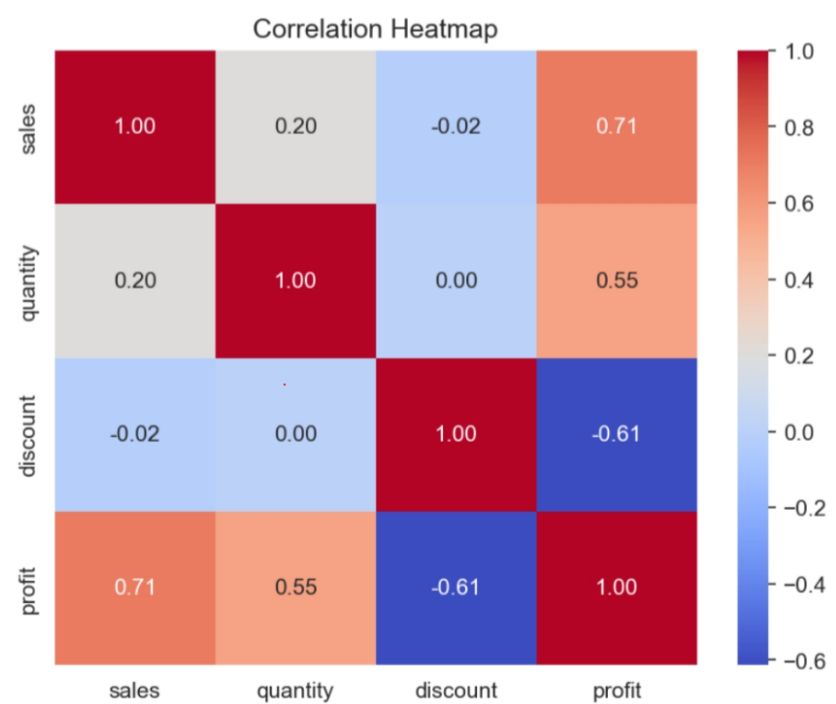
The histogram shows the distribution of profit values. The dashed vertical line marks the break-even point at profit = 0.

Average Profit by Sub-Category



Negative average profit values are highlighted in red to draw attention to weaker product lines.

Correlation Heatmap



The heatmap summarises the relationships between sales, quantity, discount, and profit.

TASK 5

5.1 several significant insights on sales performance, profitability and discount tactics were conveyed by analysing the superstore dataset using EDA and the interactive dashboard.

First off, since every transaction had non-negative profit numbers, the subcategory analysis did not identify any products that were losing money. Nonetheless, a number of subcategories showed poor profitability and many had average point margins that were either absent or unclear indicating possible problems with data quality. Only a few categories like accessories displayed quantifiable positive margins (about 21.46%) while the other categories lacked adequate acceptable data for assessment, according to the dashboard's average profit by subcategory chart. This suggests that some product lines should still be examined because the profitability data is either insufficient or untrustworthy.

Second, geographical study revealed that various portions performed differently. The regional performance chart on the dashboard revealed that certain regions like the East and Central regions contributed little to no profit despite having respectable sales figures. This points to inefficiencies in cost control, pricing or discounting tactics in particular areas and market niches.

The significant inverse link between discount and profit is a crucial finding from both the dashboard and the EDA. Higher discounts considerably lower profitability according to my correlation research which revealed a value of roughly -0.61. This is further supported by the sales vs profit scatter plot which shows that transactions with larger discounts typically cluster around lower profit values. This implies that discounting above a particular point has a detrimental effect on total profit. Discounts roughly over 0.3 to 0.5 seem to be linked to worse profitability while a specific threshold cannot be established.

The monthly sales trend chart also showed variations in sales over time with distinct peaks in some months suggesting possible seasonality. All things considered, the dashboard offered a thorough grasp of the relationships between sales, profits and discounts allowing a deeper comprehension of performance trends across geographical areas, product categories and clientele.

5.2 several practical suggestions for enhancing the businesses performance can be offered in liht of the dashboards results and EDA.first,since the data showed a significant negative asscoiation about -0.61 between discount and profit,the company should optimise its dicount approach.higher discounts are regularlynlinked to worse profitability as demonstrated by the sales vs profit scatter plot. in order to avoid severe margin erosion,it i advised that discounts be kept to less than 30% wherever feasible. overall profitability is predicted to increase by roughly 10% to 20% as a result. second,the business should concentrate on enhancing profit tracking and data quality. the lack of transactions that result in a loss and the existence of missing profit values point to possible problems with thr way profit is reported. more dependable analysis and improved decision making will be made possible by carrying out a data audit and guaranteeing precise profit computations. finally,the business should focus on underperfomring areas and clientele. the dashbaord reveals that some areas produce sales but no profit which suggests inefficiencies in cost control or pricing. the business can boost productivity and possibly raise regional profitability by 5 to 15% by examining pricing olicies and operational procedures in these areas.

5.3 the lack of negative profit values which defies the anticipated business scenario is one shortcoming of this approach.this makes it more difficult to pinpoint products that are losing money. the absence of cost breakdown data is another drawback that hinders a more thorough examination of the reasons behind some transactions lower profitability. other features for upcoming projects could be using cost and operational expense information to have a deeper undertsanding of profit drivers and to enhance segmentation tactics include customer level analysis such as lifetime value and repeat purchases. the accuracy and complteness of the profit values in the dataset was a crucial assumption made during the research.however,this assumption might not be true given the lack of negative earnings which could compromise the validity of conclusions.

REFLECTION FOR TASK 1 TO TASK 4

task 1 - i had trouble with importing the dataset because of an encoding problem (UnicodeDecodeError) which showed that the file was not in UTF-8 format. i fixed this by reading the csv file with encoding ='latin1' which made it possible for the load to load correctly. this stage emphasised how crucial it is to comprehend data formats prior to analysis. during my preliminary investigation, i noticed that the dataset had both numerical and category characteristics and that both positive and negative values were anticipated in the profit column. this discovery helped me formulate my business inquiries especially those related to understanding profitability patterns and recognising products that are losing money.

task 2 - this task posed the greatest difficulties. one significant problem i experienced was a KeyError: 'profit' which resulted from inconsistent column naming including space and case variations. in order to fix this, i standardised all column names by converting them to lowercase and removing spaces using string operations. because numeric fields like sales and profit were saved as strings, i also encountered a TypeError when calculating profit margin. this was fixed by using pd.to_numeric() with error handling to convert these columns to numeric. since null values made up over 25% of the dataset, managing missing values was a crucial data cleaning choice. to prevent losing a sizable amount of data, i decided to save these rows and handle missing values carefully during analysis rather than discarding them. i also found that several rows needed to be carefully cleaned and converted since they were incorrectly formatted which explained previous parsing errors.

task 3 - managing missing values was an important data cleansing decision because null values accounted for more than 25% of the data set. i chose to keep those rows and carefully handle missing values throughout the analysis rather than eliminating them in order to avoid losing a significant portion of the data. additionally, i discovered that a number of rows required meticulous cleaning and conversion due to improper formatting which clarified earlier the parsing errors.

task 4 - i had trouble incorporating matplotlib visualisations into the dash program during task 4. in particular,matplotlib was utilizing a gui backend that was incompatible with dash which resulted in a runtime issue due to tkinter (main thread is not in main loop) . i fixed this by using the non-gui 'agg' backend which enables proper generation of static pictures. another problem was that non-string values in category column caused a TypeError to be generated when visualising subcategory data. before graphing,the data was cleaned and values were converted to strings. my comprehension of the dataset was greatly enhanced by the dashboards interactive feature which lets me dynamically filter data and see correlations between factors like sales,profits and client groups. the dashbaord provided a more accurate depcition of how data is used in business decision making than static analysis making it simpler to spot patterns,trends and performance variations across regions and categories.

MANDATORY REFLECTION

1. during the analysis , i came across the following particular error 'profit' is the key error.this occurred when i attempted to use profit column to calculate profit margin but pandas was unable to locate the precise column name. i started by printing names of every column in the dataset to check how they were stored in order to fix it.i saw tht the olumn names irregular formatting including variations in case and spacing was the root problem. i discovered that the following specific issue during analysis 'profit' is the crucial errorr.

2.how to deal with the datasets missing values was a crucial data cleaning choice.i had to decide whether to remove or maintain the rows that included null values across several columns in about 25% of the records. eliminating them would have limited the quatity of data accessible for analysis and eliminated a significant percentage of the dataset.however,imputing values across a variety of business related fields including profit,category and region may lead to false assumptions.i made the decision to retain those rows in the dataset and where necessary use filtering techniques like dropna() to handle missing values selectively during particular studies.

3. a subcategory profit chart was one of the first visualisations i made. but , i ran across problems with matplotlib plotting during implementation because of missing and non-string values in the subcategory column.furthermore,variations in profitability between subcategories were not made evident in the original form.i initially cleaned the data,eliminated any missing values and converted the subcategory column to string format in order to better the visualisation.the chart was redrawn as a horizontal bar graph with negative or weaker values emphasized in red,arranged from lowest to highest average profit. becuae it fixed the technical issues,enhanced readability and made it simpler to spot underperforming product subcategories,all of which are in line wiith the business goal of evaluating profitability.this final version was more suitable.

4.the datasets lack of loss making transactions that is total loss of 0.00 and no rows with profit values below zero was the most unexpected result of my investigation.i was somewhat taken aback by this as the business context made it clear that negative profit values were

significant and definitely to be looked into. I would often anticipate that at least some transactions in a retail dataset would be unproductive, particularly when there are significant discounts. This result raised the possibility that some negative profits were lost during preprocessing or that there were missing profit values in the cleaned dataset. It was notable because it made it more difficult to adequately address the business question about subcategories that were losing money.

5. The one thing I would do differently is to thoroughly validate the dataset before beginning the cleaning and analysis procedure. Before moving further with feature creation and visualisations, I would specifically make sure that important columns like profit still contained the expected range of values, including negative profits. I would take this action since a number of my subsequent conclusions were impacted by problems with the quality of the data, particularly the missing numbers and the lack of transactions that resulted in losses. In order to increase confidence in the study and guarantee that the final results supported by solid evidence, a more comprehensive validation process at the outset would be beneficial.

I declare that this assignment is my own work. I did not use AI tools. I understand that I am responsible for understanding all code submitted and can explain any part of my analysis upon request.

References

richfield graduate institute of technology. (2026). *DATA MANIPULATION AND VISUALISATION 600* .
durban: richfield graduate institute of technology.